ModelIt!

Student Activity Pack

Where Did the Puddle Go?

How Heating and Cooling Change Matter

Grade 2 | NGSS 2-PS1-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever noticed a puddle after rain? What happened to it later?**

2. **Where do you think the water goes when a puddle dries up?**

3. **Draw what you think happens to puddle water on a hot sunny day.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sun Heat | Evaporation Speed | Puddle Size

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sunny Day

Sun Heat set to high

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cloudy Day

Sun Heat set to low

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Partly Cloudy

Sun Heat set to medium

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wind Speed — How fast the air is moving over the puddle — wind helps carry water vapor away faster

Puddle Depth — How deep the puddle is at the start — deeper puddles take longer to evaporate

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

When a puddle disappears on a sunny day, where does the water actually go? Use evidence from your simulation to support your answers.

1 Where is the puddle water right now if it is not on the ground anymore?

2 How is what happened to the puddle similar to what happens when you breathe on a cold window?

8 STEM Challenge: Build a Mini Water Cycle

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YOUR ENGINEERING MISSION

Design a sealed container that shows evaporation and condensation happening — your own mini water cycle!

THE SCENARIO

A science museum wants a simple display that shows kids where puddles go. Your team will build a mini water cycle in a sealed container so visitors can watch water evaporate and condense over and over again.

Engineering Design Thinking

1. WHAT KIND OF SEALED CONTAINER WILL YOU USE — A BAG, A JAR, OR A PLASTIC BOX?
2. HOW WILL YOU ADD HEAT — SUNLIGHT, A LAMP, OR WARM WATER?
3. WHERE WILL YOU LOOK FOR CONDENSATION — ON THE LID, ON THE SIDES, OR SOMEWHERE ELSE?
4. HOW WILL YOU PROVE THAT THE WATER THAT EVAPORATED IS THE SAME WATER THAT CONDENSED?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Meteorologists study weather and predict when it will rain, snow, or be sunny. They use science about evaporation and condensation every day. They earn $50,000–$95,000/year.